Effect of Glycemic Control on Early Diabetic Renal Lesions

A 5-Year Randomized Controlled Clinical Trial of Insulin-Dependent Diabetic Kidney Transplant Recipients

Jose Barbosa, MD; Michael W. Steffes, MD, PhD; David E. R. Sutherland, MD, PhD; John E. Connett, PhD; K. Venkateswara Rao, MD; S. Michael Mauer, MD

Objective.—To determine whether optimized glycemic control in type I diabetic recipients of renal allografts will prevent or delay diabetic renal lesions in the allograft.

Design.—Prospective, controlled, and randomized trial of glycemic control in an inception cohort of type I diabetic renal allograft recipients. The experimental group underwent maximized glycemic control, and the standard group was treated in the same way as other patients in the transplant clinic. Patients underwent baseline (before transplant) and 5-year posttransplant allograft biopsies.

Setting.—University of Minnesota Hospital and Clinic and the Clinical Research Center and Hennepin County Medical Center, Minneapolis.

Patients.—Type I diabetics with terminal diabetic renal failure undergoing renal transplantation. Forty-eight patients randomized to maximized or standard control completed the trial.

Intervention.—Subcutaneous insulin given several times a day or continuously (maximized group) and once or twice each day (standard group) was used throughout the trial. A significant difference for hemoglobin A_1 level was maintained (mean±SD: standard, 0.117±0.013; maximized, 0.096±0.016; $P<.001$).

Main Outcome.—The primary end point of this trial was the difference between the groups in renal glomerular mesangial expansion as determined by electron microscopy.

Results.—There was a more than twofold increase in the volume fraction of mesangial matrix per glomerulus in the standard group (mean±SD, 0.043±0.034) compared with the maximized group (0.019±0.038; $P=.024$). The threefold increase in arteriolar hyalinosis, the greater widening of the glomerular basement membrane, and increase of volume fraction of the total mesangium in the patients who received standard treatment all approached significance ($P=.10$ or less). The incidence of severe hypoglycemic episodes was greater in the maximized group (1.7 per patient per year) than in the standard treatment group (<0.1 per patient per year; $P<.001$).

Conclusions.—This trial indicates a causal relationship between hyperglycemia and an important lesion of diabetic nephropathy, mesangial matrix expansion, in renal allografts transplanted into diabetic recipients. In addition, the results with other lesions central to the development of diabetic nephropathy support the major conclusion.

(*JAMA*. 1994;272:600-606)

From the Departments of Medicine (Division of Endocrinology and Metabolism) (Dr Barbosa), Laboratory Medicine and Pathology (Dr Steffes), Surgery (Dr Sutherland), and Pediatrics (Dr Mauer), University of Minnesota Medical School, University of Minnesota Hospital, Minneapolis; Division of Biostatistics, School of Public Health (Dr Connett) and Hennepin County Medical Center (Dr Rao), University of Minnesota.

Reprint requests to Box 716, University of Minnesota Hospital and Clinic, 420 Delaware St SE, Minneapolis, MN 55455 (Dr Barbosa).

DIABETIC glomerulopathy, first reported by Kimmelstiel and Wilson in 1936,[1] ultimately results in overt renal disease in 25% to 40% of type I diabetic patients. Once overt proteinuria develops, there is a substantial risk of death from renal failure and from cardiovascular events.[2] Using morphometric analysis of biopsy specimens from renal allografts, this study was designed to determine whether the microangiopathic complications of diabetes can be prevented by improved control of blood glucose level.[3-5]

Experiments with animal models of diabetes have demonstrated that normoglycemia prevents or ameliorates microangiopathic lesions, including glomerular abnormalities.[6,7] In dogs, it was suggested that primary intervention, ie, normalization of hyperglycemia from the time of induction of diabetes, was more effective in preventing diabetic retinopathy than was secondary intervention, ie, normalization of hyperglycemia 2 years after diabetes induction.[8] Large observational cohort studies in humans have also suggested that hyperglycemia is associated with increased risk of microvascular complications, and that this relationship may be causal.[9,10] However, these retrospective analyses could not determine whether the greater tendency to develop complications in patients with worse glycemic control resulted from the glycemia per se or some covariable related to the difficulty to achieve control.

With measurements of glycosylated hemoglobin and ambulatory blood glucose levels appearing in the 1970s, it became possible to perform relatively accurate longitudinal assessment of glycemic control in diabetic patients. Since then, several prospective clinical trials of the effect of intensive vs standard glycemic control on renal function have been conducted.[11-16] Recently, the results of the Diabetes Control and Complications Trial demonstrated the beneficial effects of intensive glycemic control on several complications of diabetes, including some of the functional manifestations of diabetic nephropathy.[16] None of these trials used changes in renal structure over time as a key study end point, or studied inception cohorts.

Downloaded From: http://jama.jamanetwork.com/ by a Simon Fraser University User on 06/12/2015

Our earlier findings of diabetic lesions emerging after only 2 to 4 years in normal kidneys transplanted into type I diabetic patients[17,18] suggested this as a practical setting to study the relationship of glycemic control to the early renal lesions of diabetes. We report herein the results of a long-term, prospective, and randomized clinical trial of type I diabetic renal allograft recipients constituting an inception cohort, ie, all patients were at stage 0 for diabetic renal lesions in the graft when randomized into the trial. This study was designed to test the hypothesis that improved glycemic control in type I diabetic patients could prevent or minimize the development of diabetic lesions in the renal allograft.

PATIENTS AND METHODS

Patients

All patients considered for the study were diagnosed as having insulin-dependent type I diabetes and had received a renal allograft as treatment for end-stage diabetic nephropathy. The study was approved by the University of Minnesota Committee on the Use of Human Subjects in Research, Minneapolis. The subjects were recruited from the large population of type I diabetics with uremia who sought kidney transplantation at the University of Minnesota. The Transplant Center had performed transplants in more than 500 diabetic patients when we began recruitment for the trial, representing a referral population from all 50 states, although approximately 90% came from the upper Midwest and all were white. About 300 patients were invited to join the study; 100 declined to join. One hundred one accepted, but for a number of reasons (including death, major illness precluding transplantation, and logistical problems of living or being treated out of state) were not randomized. The remaining 99 were originally randomized equally between the two treatment groups (Table 1). Eighty-five percent came from Minnesota and surrounding states, and all were white. When the study began in 1978, patients were randomized to maximized or standard glycemic control before transplantation. Two years later, we began performing randomization 3 months after transplantation to ensure that an adequate intraoperative baseline renal allograft biopsy specimen had been obtained and to exclude patients whose grafts were rejected during this high-risk period. Between 1978 and 1984, randomization was 2:1 in favor of the maximized group, since we hypothesized that this group would have more withdrawals from the trial. This was not the case.

Table 1.—Profile of Trial Participants and Biopsy Specimens Completed*

	Maximized Treatment	Standardized Treatment	P
Patients randomized, No.	52	47	...
Sex, No. M/F	41/11	34/13	.60
Age, y	35±6 (21-50)†	36±8 (21-58)	.48
Duration of diabetes, y	23±6 (14-39)	21±5 (14-36)	.08
Patients who became unavailable for the study, No.	27	24	...
Sex, No. M/F	19/8	17/7	.79
Age, y	34±7 (21-50)	37±9 (22-58)	.69
Duration of diabetes, y	22±6 (14-39)	21±4 (14-30)	.49
Deaths, No.	7	8	.79
Graft losses/chronic rejection, No.	2	2	.70
Voluntary withdrawals before final biopsy, No.	9	4	.30
Inadequate or no tissue from baseline biopsy, No.	7	6	.80
Inadequate tissue from 5-y biopsy, No.	2	4	.56
Patients who completed the study, No.	25	23	...
Sex, No. M/F	22/3	17/6	.38
Age, y	36±5 (27-50)	35±7 (21-49)	.57
Duration of diabetes, y	24±6 (16-38)	22±5 (14-36)	.22
Body mass index	28±8.2 (19-57)	26±4.3 (21-49)	.30
Blood pressure			
Systolic			
Entry	132±9 (120-140)	126±10 (101-142)	.03
During trial	131±7 (115-143)	129±9 (112-139)	.39
Diastolic			
Entry	78±4.9 (66-85)	78±5.2 (68-87)	.99
During trial	77±6 (61-83)	75±8 (67-86)	.88
Insulin dose during study, U/d	77±29 (31-168)	49±19 (20-81)	<.001

*Values for age, duration of diabetes, body mass index, and blood pressure are from the time of randomization.
†Mean±SD (range).

To balance the size of the two groups, in 1985 we began 2:1 randomization in favor of the standard group. Analyses of end point data were accordingly stratified by time period, as described below.

In addition to the baseline needle biopsy performed at transplantation, patients underwent percutaneous allograft needle biopsies 5 years after transplantation in the Clinical Research Center at the University of Minnesota Hospital or at Hennepin County Medical Center, Minneapolis. Patients participated in the trial for 5 years, with a window for biopsy extending from 4½ to 5½ years. There was no difference between the treatment groups for the time of participation. The demographic and clinical characteristics of the patients who completed the trial were similar in all subjects originally randomized (Table 1). Of the patients who completed the trial, the maximized group had a slightly, but significantly, higher mean systolic blood pressure at the time of randomization. However, there were no differences between the two groups in mean systolic or diastolic blood pressures during the 5 years of the trial (Table 1).

Of those subjects who completed the trial, grafts were obtained from either living related donors (LRDs) (36 patients, of whom 16 were randomized to maximized treatment and 20 to standard treatment) or cadaver donors (CADs) (12 patients, of whom nine were randomized to maximized treatment and three to standard treatment). Most LRDs were siblings (78%); 14% were parents; and 8% were other relatives. Siblings had been discordant for diabetes with the recipient for 20 years or longer; therefore, their risk of developing diabetes was considered low. All LRDs had normal glucose tolerance, and none later became diabetic. All CADs had negative histories for diabetes.

Treatment

Standard Treatment.—The management of glycemia in these patients was method oriented and included the goal of avoiding hyperglycemic symptoms. Since 1980, we tried to avoid hemoglobin A_1 (HbA_1) values over 0.12. This value was selected after we determined the mean HbA_1 level of 43 randomly chosen diabetic kidney recipients in our transplant clinic to be 0.119. Several times during the trial, the HbA_1 level of the patients who received standard treatment was compared with that of a random sample of approximately 40 diabetic kidney transplant recipients not participating in the trial. There were no differences in HbA_1 levels between the two groups.

Patients in the standard treatment group received one daily injection of a mixture of isophane insulin and crystal-

Downloaded From: http://jama.jamanetwork.com/ by a Simon Fraser University User on 06/12/2015

line regular insulin (100 U/mL, beef/pork) 30 minutes before breakfast. Approximately one third of the patients in the standard group were treated with two injections of a mixture of isophane insulin and crystalline regular insulin daily for extended periods to alleviate hyperglycemic symptoms or reduce HbA_1 values to 0.12 or below. Periodically, an effort was made to return to one daily injection, as long as the above criteria were met. During the last 5 years of the study, most patients' drug regimen was changed to human insulin.

Earlier in the trial, patients limited their self-monitoring to urine testing one to four times daily. As home blood glucose monitoring had become common practice in our medical community by 1987, most patients in the standard group started using this technique once or twice daily, albeit without appreciable influence on metabolic control. The patients in the standard group were generally contacted by the trial nurse every 2 to 4 months (more often if there were concerns, such as symptomatic hyperglycemia, HbA_1 level >0.12, or hypoglycemic episodes) and by a dietitian once a year. However, the patients could contact the dietitian as often as they desired. The dietitian recommended an American Diabetes Association weight-maintaining diet. For patients more than 15% above ideal body weight, a weight-reducing diet was recommended. The study nurse recommended exercise in general terms.

Maximized Treatment.—The approach was goal oriented, with the insulin regimen adjusted to each patient's needs and preferences, which often varied during the 5 years. The goal was to obtain HbA_1 values within normal limits. The nurse clinician contacted patients two to three times weekly. Patients were asked to monitor their home blood glucose level four to six times daily, allowing frequent readjustment of the insulin dose, diet, and exercise to meet the study goals. This intensive supervision was considered more important than the insulin type or the method of insulin delivery.[19] Approximately 30% of these patients used continuous subcutaneous insulin infusions during the early and mid 1980s.[19] Others chose multiple daily injections, most commonly consisting of a mixture of beef/pork isophane insulin and crystalline regular insulin before breakfast, crystalline regular insulin before supper, and isophane insulin at bedtime. Some patients also took crystalline regular insulin at lunch, and a few were treated with extended insulin zinc once or twice a day and crystalline regular insulin at the three main meals. A few patients achieved a normal HbA_1 level with injections of a mixture of isophane insulin and crystalline regular insulin twice a day. Some patients in the maximized group opted to change their insulin regimen, under our supervision, several times during the 5 years. In the late 1980s, most switched to multiple injections, but differences in glycemic control were not noted. During the last 5 years of the trial, many patients used human insulin. The patients in the maximized group were contacted frequently by the study dietitian. An American Diabetes Association weight-maintaining or weight-reducing diet was recommended, and a constant effort was made to adjust the diet to the patient's lifestyle, preferences, and current insulin regimen. Change in exercise was also frequently tried to achieve better and safer glycemic control. Precautions were taken to minimize hypoglycemic episodes and their consequences. In addition to close supervision by the study personnel, family members were periodically trained in the use of glucagon.

Immunosuppression and Other Treatment

Immunosuppression regimens used during the period of the study (1978 through 1992) have been summarized elsewhere.[20] The immunosuppressive protocols and drug dosages used did not differ between the two groups. Most diabetic patients had hypertension and were treated vigorously. Their blood pressure was measured frequently, with the results monitored by study personnel. There were no differences in the antihypertensive drugs used in the two experimental groups. Angiotensin-converting enzyme inhibitors were used infrequently and only in the last 2 to 3 years of the study. Patients were rarely hypertensive during the trial, and there were no blood pressure differences between the groups during the course of the study.

Hypoglycemic Episodes

Hypoglycemic episodes, classified according to severity of symptoms, were monitored throughout the study. Severe hypoglycemia was defined as an episode of behavioral change requiring the help of others for treatment. Moderately severe hypoglycemia included severe hypoglycemic symptoms but with preserved capability for self-treatment. Most patients participating in the trial had good family support, and all had excellent nursing interaction, thus lessening the risks of severe hypoglycemia. Most episodes were treated at home. There were 29 visits to hospitals for treatment of severe hypoglycemia, almost all (26 of 29) in the maximized group. Only one patient (in the maximized group) was hospitalized (for 2 weeks), remaining in a coma for 6 days. Fortunately, this patient recovered his mental function completely. There were no automobile or other accidents during the trial, and there were no deaths from hypoglycemia.

Patients Who Left the Study

Patients left this trial for many reasons, with equal numbers between treatment groups (Table 1). There were 15 deaths, equally distributed between treatment groups (Table 1). Seven were from cardiovascular disease, three from septicemia, four from complications of uremia, and one from suicide. The causes of death were similar in both groups, and all the patients (except the victim of suicide) died in a hospital. No death was related to treatment of diabetes, ie, hypoglycemia or hyperglycemia.

Laboratory Studies

Fasting serum glucose, creatinine, urea nitrogen, and HbA_1 levels were measured every 3 months in the laboratories of the University of Minnesota Hospital, in addition to home capillary blood glucose measurements.[19] The HbA_1 measurements were made initially by ion-exchange columns and from 1986 by high-performance liquid chromatography (BioRad Laboratories, Hercules, Calif). Values are expressed as HbA_1 levels, since the earlier procedure only measured HbA_1. The normal range was 0.051 to 0.073. Interassay coefficients of variation ranged from 4% to 5% with the early method to 2% to 3% with high-performance liquid chromatography.

Renal Biopsies

Patients underwent kidney biopsies during transplant surgery and 5 years later. In LRDs, the baseline biopsies were performed with 8.5-cm, 16-gauge disposable (Trucut, Baxter Co, Valencia, Calif) or reusable needles (Vim-Silverman, Baxter Co) before removal of the kidney from the donor. Biopsy specimens were obtained from the CAD kidneys with a needle after graft revascularization in the recipient, with bleeding being controlled with local pressure. There were no complications from this procedure. The fifth-year biopsies were done percutaneously with a needle (Vim-Silverman) and examined immediately with a dissecting microscope to ensure adequacy of the sample and correct division of the specimen for the histologic studies described below. We previously documented that morphometric values were not different in biopsy specimens of normal LRD vs CAD kidneys obtained

Downloaded From: http://jama.jamanetwork.com/ by a Simon Fraser University User on 06/12/2015

at the time of transplantation.[21]

Light Microscopy.—Part of the biopsy specimen was placed in Zenker's solution and prepared for light microscopy, as previously described.[17] Periodic acid-Schiff–stained sections were coded and evaluated without knowledge as to source. On the basis of this evaluation, patients were excluded from the study if they had transplant glomerulopathy with mesangial interposition as defined by periodic acid-Schiff–positive double contours of the glomerular capillary walls.[22] The suspected presence of acute or chronic transplant rejection or cyclosporine toxic effects was not, per se, reason for termination from the study. However, results of biopsy specimens obtained during episodes of acute rejection were not included. Lesions of arteriolar hyalinosis were graded in a masked fashion as previously described.[23] Briefly, the index of arteriolar hyalinosis was determined as a semiquantitative estimate of replacement of arteriolar smooth muscle by waxy, homogeneous, periodic acid-Schiff–positive material. Incomplete replacement of a few vessels was graded as 0.5; incomplete replacement of most vessels was graded as 1.5; complete replacement of approximately one half of the arterioles was graded as 2.5; and complete replacement of most of the arterioles was graded as 3.5.

Electron Microscopy.—Part of each biopsy specimen was fixed in 1% or 2.5% glutaraldehyde in 0.1-mol/L cacodylate buffer. No differences in glomerular structure were found in pieces of the same tissue separately fixed with these two different concentrations of fixative. Tissues were postfixed in osmium tetroxide and processed with embedding media (Polybed 812, Polysciences, Warrington, Pa) for thin sectioning.

Morphometric Analyses.—Poorly fixed biopsy specimens were not measured. We tried to measure six (5.6±1.1) nonsclerotic glomeruli per biopsy specimen. The centermost glomerulus from each block was sectioned except when glomeruli were found in only one or two blocks. Systematic procurement of electron micrographs (approximately ×3900 magnification) of each glomerulus provided photographs for the assembly of a montage of the entire glomerular cross section for the measurement of mesangial volume fraction (Vvmes/glom). The mesangial limits were defined by the loss of parallelism between endothelial and epithelial cell cytoplasm in the glomerular capillary wall.[23,24] Glomerular basement membrane (GBM) lying between the mesangium and epithelial cells was included in this measure. In addition, systematic unbiased photographs of the glomerulus were taken at ×11 000 (representing approximately 15% to 20% of the glomerular cross section) for the measurement of GBM width, mesangial matrix volume fraction (Vvmat/glom), and mesangial cell volume fraction (Vvcell/glom). The mesangial GBM was excluded from the measure of mesangial matrix volume fraction.[23,24]

The GBM width was obtained by the orthogonal intercept method and was calculated as the harmonic mean of at least 100 measurements per biopsy specimen.[25]

Mesangial volume fraction was estimated by point counting[23,24] on the low-magnification montages. A double-lattice grid consisting of coarse points (CPs) (6 mm apart) and fine points (FPs) (3 mm apart) was placed over the montage. There were four FPs for each CP. The following equation was used: Vvmes/glom=FPs on the mesangium/(CPs on the glomerulus × 4).

Mesangial cell volume fraction and mesangial matrix volume fraction were estimated by placing a grid of points 3 mm apart over the high-magnification micrographs. The points falling on mesangial cells (MC), mesangial matrix (MM), and mesangial GBM (MGBM) were noted.[26] The volume fractions of mesangium occupied by mesangial cells and mesangial matrix were calculated by the following equations:

VvMC=MC/(MC + MM + MGBM);
VvMM=MM/(MC + MM + MGBM);
Vvcell/glom=VvMC×Vvmes/glom; and
Vvmat/glom=VvMM×Vvmes/glom.

Clinical Trial Statistics

A sequential monitoring plan was developed for this clinical trial as a guide for possible early termination if a pattern of strong differences between the treatment groups emerged before completion of randomization and follow-up. Monitoring of interim data for this study was complicated for a number of reasons, including the following.

1. Multiple end points were being studied; GBM widening, mesangial expansion, and changes in mesangial matrix were all considered important. Early in the study, we decided that the primary measured end point for decision making to continue or stop the trial would be mesangial expansion as determined by electron microscopy.

2. End points in some patients were unobservable because of death or rejection of kidney transplants, or because patients refused to undergo follow-up biopsies. In general the patients were at high risk of death, and transplant rejection was not unlikely.

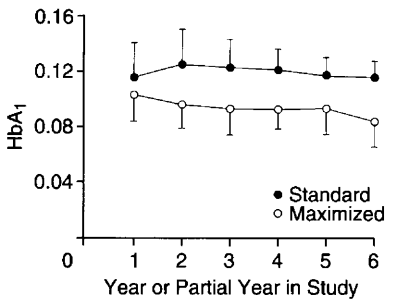
Monitoring analyses were conducted at 6-month intervals by two of us (M.W.S. and J.E.C.) who were not involved in patient care. The results were not disclosed to other investigators, nurses, or other personnel. The desired overall level for significance testing was α=.05 (two sided). A complex simulation study was carried out to determine a nominal significance level adjusting for multiple looks at the data, based on sequential Mann-Whitney rank-sum tests, assuming that the end point was (1) electron microscopic mesangial expansion only or (2) electron microscopic mesangial expansion or death (with death considered a treatment failure regardless of cause and assigned an arbitrarily high value in the rank-sum analyses). The simulation study indicated that, for end point 1, the nominal significance level should be .024, whereas for the combined end point 2, the normal significance level should be .015. That is, to observe a difference in the treatment groups for end point 1 that would exceed a significance level of .05 in the course of sequential monitoring, one would have to observe a difference in one of the sequential analyses that had a nominal significance level of .024 or less. At no time during the operational period of the trial were nominal significance levels for the electron microscopic mesangial expansion end point or for the combined death–electron microscopic mesangial expansion end point below the cutoffs specified above.

Stratified analyses, to control for the two randomization schedules, were performed by means of Fisher's method for combining the results of independent tests.[27,28] The P values presented were all determined by this procedure as applied to the Mann-Whitney test.

Additional simulation analyses were conducted to estimate the power of the trial to detect clinically significant treatment effects, given the numbers of patients expected to be randomized and evaluable. If, for example, mesangial expansion in patients in the maximized control group was 50% of that for those in the standard control group, the estimated statistical power was 87% for end point 1 and 45% for end point 2. This was based on the expectation that 80 patients would be randomized.

The actual numbers of patients available for analysis of 5-year electron microscopic mesangial expansion were 25 in the maximized treatment group and 23 in the standard treatment group. Given the observed electron microscopic mesangial expansion in the standard group and the observed within-group variance of this end point, a post hoc estimate of the statistical power for detecting a mean 50% reduction in electron microscopic mesangial expansion

Downloaded From: http://jama.jamanetwork.com/ by a Simon Fraser University User on 06/12/2015



Hemoglobin A_1 (HbA_1) levels in patients who received standard and maximized treatment for the duration of the trial. Values are for each full or partial patient year in the study and are expressed as mean±1 SD. They demonstrate clear and consistent differences between groups (P<.001 for the duration of the trial).

in the maximized group may be computed. We found this post hoc estimated power to be 33%. Thus, primarily because the number of evaluable patients was small, it is evident that the power of this trial to detect a clinically important treatment effect was not high.

Although sequential monitoring for the study was carried out with adjustment for multiple looks at the end point data, all significance levels in the present report are *not adjusted* for multiple comparisons. That is, only nominal significance levels based on the completed follow-up data set are presented. Multiple comparison adjustments, such as those of Bonferroni, Tukey, Newman, and others,[29] have the merit of keeping the studywide type I error rate at the desired level but also can greatly increase the type II error rate. The reader may choose to make a conservative Bonferroni adjustment by multiplying the nominal P values by the number of between-group comparisons in study end points.

In all instances, Mann-Whitney tests, combined across strata by Fisher's method (see above), were used to determine differences between treatment groups. All values are expressed as mean±SD.

RESULTS

As measured by HbA_1, the treatment groups maintained a difference in glycemic control for the duration of the trial (standard, 0.117±0.013; maximized, 0.096±0.016; P<.001; Figure), but most patients had values exceeding the normal range of 0.051 to 0.073. There were no significant differences in systolic or diastolic blood pressures in the two treatment groups during the 5 years of the study (Table 1). The values for the morphometric measures at baseline and 5 years are presented in Table 2, and the differences between baseline and 5 years in Table 3. The increase in GBM width tended to be less in the maximized group, with a P value that approached significance (Table 3). Similarly, the increase in mesangial volume fraction tended to be less in the maximized group; there was no difference in mesangial cell volume fraction. The increase in mesangial matrix volume fraction was more than two times greater in the standard group (P=.024, Table 3). The index of arteriolar hyalinosis tended to be greater in the standard group (0.6±0.7, n=20) than in the maximized group (0.2±0.4, n=25) (P=.086). The incidence of severe hypoglycemic episodes was greater in the maximized group (1.7 per patient per year) than in the standard treatment group (<0.1 per patient per year) (P<.001).

COMMENT

Diabetic patients randomized to maximized treatment had a smaller accumulation of mesangial matrix at 5 years after renal transplantation than did patients randomized to standard treatment. Thus, this study documents that improved glycemic control retards the development of critical diabetic glomerular lesions. Furthermore, the trends for GBM width, volume fraction of the total mesangium, and arteriolar hyalinosis all suggested further benefits of improved glycemic control. These results suggest a causal relationship between hyperglycemia and mesangial expansion, especially mesangial matrix accumulation, the dominant component in mesangial expansion in type I diabetic patients.[26] In addition, the confluence of the direction of all results indicates a similar trend among other important glomerular and vascular lesions that characterize the pathologic findings in diabetic nephropathy.

Conclusions of our study may be tempered somewhat by the relatively high proportion of patients for whom data could not be evaluated. Major reasons were death, voluntary withdrawal before the final biopsy, and inadequate tissue from the baseline biopsy. The last cause is of less potential concern than the others, since the failure to obtain baseline biopsy specimens occurred before randomization and could not have been affected by the treatment. The two treatment groups were approximately balanced in percentages of inadequate baseline biopsy specimens (maximized group, 13.5%; standard group, 12.8%), deaths (maximized group, 13.5%; standard group, 17.0%), and graft losses or transplant glomerulopathy (maximized group, 3.8%; standard group, 4.3%). Two patients in the maximized group and four in the standard group had inadequate 5-year biopsy specimens (3.8% vs 8.5%; P=.42). The rate of voluntary withdrawal tended to be higher in the maximized than the standard group (17.3% vs 8.5%, respectively; P=.24), perhaps because rigorous behavior modifications were expected in the maximized treatment group. Although the dropout rate was high, there was no obvious source of bias that would affect our conclusions regarding maximized vs standard glycemic control. Influences of other factors, such as blood pressure, diet, type of immunosuppression, or duration of disease, on the results reported herein are unlikely. Duration of disease, blood pressure, and immunosuppressive treatment during the study were not different between the two groups. Although this study did not quantitate dietary intake, the study personnel did not detect differences in dietary practice between the two groups by means of customary dietary recall methods. Extreme increases in dietary protein (50% protein intake) have been associated with a slightly increased mesangial fractional volume in rats with long-term streptozocin-induced diabetes compared with diabetic rats on a 20% protein diet.[30] However, this finding was not confirmed in a similar study.[31] Thus, at best, extreme dietary changes in rats could produce subtle effects on diabetic glomerular lesions. There are no data to support the hypothesis that relatively small dietary differences, as might have been missed in the present study, could influence the development of diabetic glomerular or vascular lesions in humans.

The study was able to show effects of improved glycemic control because of certain design characteristics. This was the first study, to our knowledge, to examine the role of glycemic control on diabetic glomerulopathy in a true primary intervention context, ie, in patients whose organs under study were normal at the start of the trial. Indeed, it is not feasible to recruit and randomize a population of patients on their first day of diabetes. The Diabetes Control and Complication Trial, designed primarily to study the effects of primary intervention on diabetic retinopathy, recruited patients after they had had overt diabetes for 1 year or longer.[4,16] Other trials have addressed important renal function measures but with secondary intervention strategies.[11-15] Clinically undetectable renal structural changes occurring very early in the diabetic state must underlie the progression of lesions to more advanced and functionally detectable renal injury. Thus, Østerby reported GBM thickening in native kidneys after only 1½ to 2½ years of clini-

Downloaded From: http://jama.jamanetwork.com/ by a Simon Fraser University User on 06/12/2015

Table 2.—Glomerular Morphometric Measures at Baseline and 5 Years*

Treatment	Time	GBM Width, nm	Volume Fraction		
			Mesangium	Mesangial Cell	Mesangial Matrix
Standard (n=23)	Baseline	327±41	0.198±0.030	0.087±0.022	0.082±0.013
	5 y	475±181	0.235±0.050	0.077±0.017	0.126±0.036
Maximized (n=25)	Baseline	339±47	0.212±0.031	0.093±0.022	0.090±0.015
	5 y	430±73	0.218±0.058	0.082±0.025	0.109±0.360

*Values are expressed as mean±SD. GBM indicates glomerular basement membrane.

Table 3.—Glomerular Structural Changes From Baseline to 5 Years*

Treatment	GBM Width, nm	Volume Fraction		
		Mesangium	Mesangial Cell	Mesangial Matrix
Standard (n=23)	148±166	0.037±0.054	−0.010±0.028	0.043±0.034
Maximized (n=25)	91±73	0.006±0.064	−0.011±0.035	0.019±0.038
P	.13	.086	.86	.024

*Values are expressed as mean±SD. Structural data are presented as the differences between the 5-year and the baseline vaules. GBM indicates glomerular basement membrane.

cal diabetes.[32] Furthermore, Engerman and Kern[8] showed that dogs with 2 years of poor glycemic control after induction of diabetes followed by 3 years of improved control develop retinal abnormalities identical to those of dogs with poor control throughout the 5 years. Retinopathy was prevented only in dogs treated with maximized control from the time of induction of diabetes.

This trial focused on differences between baseline and follow-up renal biopsy specimens by means of key glomerular structural end points related to mesangial expansion that were quantitated in a systematic and unbiased manner. We selected mesangial expansion as the primary end point of the trial, because we and others have previously demonstrated that mesangial expansion is the glomerular lesion of diabetes that best correlates with the clinical manifestations of diabetic nephropathy—proteinuria, declining glomerular filtration rate, and hypertension.[23,33] Moreover, we recently documented that expansion of the mesangium in diabetes primarily results from the accumulation of mesangial matrix material with a smaller contribution from the expansion of mesangial cell volume.[26] These reports highlight the importance of the observations of a salutary effect of improved glycemic control on mesangial matrix expansion in the current study.

The observation suggesting less severe arteriolar hyalinosis lesions in the maximized group supports other work indicating the importance of this structural change as a diabetic vascular lesion.[17] Recently, we showed that arteriolar hyalinosis is linked to global sclerosis and through this mechanism might contribute to declining glomerular filtration rate in type I diabetic patients with overt nephropathy.[34] Thus, the possible beneficial effect of improved glycemic control on these vascular lesions may have important clinical implications.

Although our results were obtained in the study of renal allografts and reflect relatively early lesions of diabetic nephropathy, there are strong reasons to believe that this study is relevant to diabetic nephropathology in native kidneys. First, we have previously shown that the diabetic renal lesions (mesangial expansion, increased GBM width, afferent and efferent arteriolar hyalinosis, and increased renal extracellular basement membrane localization of plasma proteins) occur in normal kidneys transplanted into diabetic patients.[17,18,35] These renal lesions do not occur in allografts transplanted into nondiabetic patients.[17,35] Second, we have shown that mesangial expansion in normal kidneys transplanted into type I diabetic patients develops at rates that parallel those occurring in the native kidneys of diabetic patients.[18] These rates ranged from unmeasurable to moderate to advanced lesions.[18] These observations support the argument that the present model is relevant to early diabetic nephropathy in native kidneys.

As suggested above, not all diabetic patients who previously experienced terminal uremia from diabetic nephropathy in their own kidneys necessarily develop progressive diabetic pathologic changes in the renal allograft. Thus, only about one third develop advanced lesions in the first 6 to 14 years after transplantation, whereas one third develop moderate lesions, and one third have little or no abnormality.[18] This proportion of patients who develop serious lesions in the renal allografts is similar to that of patients expected to develop advanced clinical diabetic nephropathy in their native kidneys.[18,35] Thus, the patients in the present study would be expected to develop renal changes in a manner similar to that of the general population of type I diabetics. It also follows, given that a substantial proportion of patients randomized to either group were likely to develop lesions slowly whatever the therapy, that observed overlap between the groups was to be expected. In this context, the significant differences demonstrated between the maximized and standard control groups, given the small number of patients studied, represents a strong indication of the power of improved glycemic control to inhibit a fundamental renal lesion of diabetes.

Nevertheless, improvement in glycemic control was not obtained without substantial risks. Despite close clinical supervision and prednisone therapy, the patients in the maximized control group had higher frequencies of moderate and severe hypoglycemic episodes than did those in the standard group.[36] Although no patient died, there were instances of life-threatening hypoglycemia. A cross-sectional study of the cognitive function in the trial patients, however, did not detect any abnormalities or differences between the two experimental groups.[37] Our patients with long-standing diabetes often had advanced neuropathy and, thus, may have been more prone to hypoglycemia than the general diabetic population.[36] The same disturbing trend was reported in the Diabetes Control and Complications Trial study of maximized glycemic control in patients with much shorter durations of diabetes and generally free of advanced diabetic complications.[4,16] Unlike the Diabetes Control and Complications Trial, where only certain patients were prone to severe hypoglycemia, most of our patients who were receiving maximized control proved vulnerable to moderate and severe hypoglycemia.[36,37] Thus, hypoglycemia remains one of the main impediments to the application of maximized glycemic control to the diabetic population at large. In a less supervised and regimented setting than that of our or other clinical trials,[4,11-16] the risks may be unacceptable. Furthermore, it remains difficult for type I diabetic patients to achieve or approach normoglycemia and to comply with currently available demanding and cumbersome treatment protocols. Improved methods of insulin delivery and glycemic monitoring will have to be developed before the knowledge derived from this and other studies can find widespread application among diabetic patients.

It is known that the increased mor-

Downloaded From: http://jama.jamanetwork.com/ by a Simon Fraser University User on 06/12/2015

tality in type I diabetes is largely confined to those patients who develop overt renal disease.[2] Thus, while the risk-benefit ratio may not favor maximized control for patients who are not developing serious renal lesions, this treatment might be appropriate for patients who are. Unfortunately, renal function studies are currently inadequate to identify individuals at risk of nephropathy until the renal lesions are advanced. This is true even when subtle functional alterations, such as microalbuminuria, reach levels that are regularly predictive of the later development of overt nephropathy.[38,39] Alternative strategies for risk identification will come only from careful longitudinal natural history studies of renal structure and function leading to the identification of predictors of clinical disease, correlated with potential genetic markers. Knowledge so derived, coupled with improved technologies of glycemic control, holds the realistic possibility of greatly diminishing the morbidity and mortality of diabetes.

This study was supported by National Institutes of Health grants AM20742, AM43605, and 2P01DK13083; National Center for Research Resources, General Clinical Research Center Program, M01RR00400; Eli Lilly Co, Indianapolis, Ind; Boehringer Mannheim Diagnostics, Indianapolis; the Minnesota Medical Foundation; and the American Diabetes Association (Minnesota Affiliate).

The dedicated contributions of Paula Knutzen, RN, Lois Menth, RN, Susan Nelson, RN, Karen Daniels, RN, Mary Jean Thompson, RN, and the nursing staff at the Clinical Research Center are gratefully acknowledged. John Basgen's unfailing technical support was invaluable. The support and advice from John Najarian, MD, statistical assistance from David Heitke, PhD, and Kevin Nepsund, the help with HLA typing from Harriet Noreen, secretarial and editorial assistance from Doug Welty, Christie Roiger, Christine Adams, and John Askelson, the help with the measurement of HbA_1 from Esther Freier, and assistance with compliance studies from John Hung, PhD, were appreciated. Early encouragement of Frederick Goetz, MD, was important at the inception of this study.

References

1. Kimmelstiel P, Wilson C. Intercapillary lesions in the glomeruli of the kidney. *Am J Pathol.* 1936; 12:83-97.
2. Anderson AR, Christiansen JS, Andersen JK, et al. Diabetic nephropathy in type 1 (insulin-dependent) diabetes: an epidemiological study. *Diabetologia.* 1983;25:496-501.
3. Winegrad A. Does a common mechanism induce the diverse complications of diabetes? *Diabetes.* 1987;36:396-406.
4. Diabetes Control and Complications Trial. Are continuing studies of metabolic control and microvascular complications in insulin-dependent diabetes mellitus justified? *N Engl J Med.* 1988;318:246-250.
5. Barbosa J. Diabetes: the science and the art. *Arch Intern Med.* 1983;143:1118-1119.
6. Mauer SM, Steffes MW, Sutherland DER, Najarian JS, Michael AF, Brown DM. Studies of the rate of regression of the glomerular lesions in diabetic rats treated with pancreatic islet transplantation. *Diabetes.* 1975;24:280-285.
7. Steffes MW, Brown DM, Basgen JM, Mauer SM. Amelioration of mesangial volume and surface alterations following islet transplantation in diabetic rats. *Diabetes.* 1980;29:509-515.
8. Engerman R, Kern T. Progression of incipient diabetic retinopathy during good glycemic control. *Diabetes.* 1987;36:808-812.
9. Pirart J. Diabetes mellitus and its degenerative complications: a prospective study of 4400 patients observed between 1947 and 1973. *Diabetes Care.* 1978;1:168-188.
10. West K, Ahuja M, Bennett P, et al. Interrelationships of microangiopathy, plasma glucose and other risk factors in 3583 diabetic patients: a multinational study. *Diabetologia.* 1982;22:412-420.
11. Deckert T, Lauritzen T, Parving H, Christiansen J, Steno Study Group. Effect of two years of strict control on kidney function in long-term insulin-dependent diabetics. *Diabet Nephropathy.* 1983;2: 6-11.
12. Kroc Collaborative Study Group. Blood glucose control and the evolution of diabetic retinopathy and albuminuria. *N Engl J Med.* 1984;311:365-372.
13. Feldt-Rasmussen B, Mathiesen E, Hegedus L, Deckert T. Kidney function during 12 months of strict metabolic control in insulin dependent diabetic patients with incipient nephropathy. *N Engl J Med.* 1986;314:665-670.
14. Hanssen K, Dahl-Jorgensen K, Laurtizen T, et al. Diabetic control and microvascular complications: the near-normoglycaemic experience. *Diabetologia.* 1986;29:677-684.
15. Dahl-Jorgensen K, Brinchmann-Hansen O, Hanssen K, et al. Effect of near normoglycaemia for two years on progression of early diabetic retinopathy, nephropathy, and neuropathy: the Oslo study. *BMJ.* 1986;293:1195-1199.
16. Diabetes Control and Complications Trial. The effect of intensive treatment of diabetes on the development and progression of long-term complications in insulin-dependent diabetes mellitus. *N Engl J Med.* 1993;329:977-986.
17. Mauer M, Barbosa J, Vernier R, et al. Development of diabetic vascular lesions in normal kidney transplanted into patients with diabetes mellitus. *N Engl J Med.* 1976;295:916-920.
18. Mauer SM, Goetz FC, McHugh LE, et al. Long-term study of normal kidneys transplanted into patients with type 1 diabetes. *Diabetes.* 1989;38: 516-523.
19. Barbosa J, Menth L, Schumacher G, et al. Feasibility of blood glucose self-monitoring in insulin dependent diabetes. *Diabetes Care.* 1980;3:155-159.
20. Fryd D, Migliori R, Simmons R, et al. Renal transplantation at the University of Minnesota during the 1980s. In: Terasaki P, ed. *Clinical Transplantation 1987.* Los Angeles, Calif: UCLA Tissue Typing Laboratory; 1987:167-181.
21. Steffes MW, Barbosa J, Basgen JM, Sutherland DER, Najarian JS, Mauer SM. Quantitative glomerular morphology of the normal human kidney. *Lab Invest.* 1983;49:82-86.
22. Habib R, Zurowska A, Hinglais N, et al. A specific lesion of the graft: allograft glomerulopathy. *Kidney Int.* 1993;44:S-104–S-111.
23. Mauer SM, Steffes MW, Ellis EN, et al. Structural-functional relationships in diabetic nephropathy. *J Clin Invest.* 1984;74:1143-1155.
24. Ellis EN, Basgen JM, Mauer SM, Steffes MW. Kidney biopsy technique and evaluation. In: Clarke WL, Larner J, Pohl SL, eds. *Methods in Diabetes Research, Volume II: Clinical Methods.* New York, NY: John Wiley & Sons Inc; 1986:633-648.
25. Jensen EB, Gundersen HG, Østerby R. Determination of membrane thickness distribution from orthogonal intercepts. *J Microsc.* 1978;115:19-23.
26. Steffes MW, Bilous RW, Sutherland DER, Mauer SM. Cell and matrix components of the glomerular mesangium in type I diabetes. *Diabetes.* 1992;41:679-684.
27. Fisher RA. Combining independent tests of significance. *Am Stat.* 1948;2:30.
28. Fisher RA. *Statistical Methods for Research Workers.* 11th ed. Edinburgh, Scotland: Oliver & Boyd; 1950.
29. Miller RG. *Simultaneous Statistical Inference.* New York, NY: Springer-Verlag; 1981.
30. Mauer SM, Steffes MW, Azar S, Brown DM. Effects of dietary protein content in streptozotocin diabetic rats. *Kidney Int.* 1989;35:48-59.
31. Rennke HG, Sandstrom D, Zatz R, Meyer TN, Gowen RS, Brenner BM. The role of dietary protein in the development of glomerular structural alterations in long-term experimental diabetes. *Kidney Int.* 1986;291:389. Abstract.
32. Østerby R. Early phases in the development of diabetic glomerulopathy. *Acta Med Scand Suppl.* 1975;574:1-82.
33. Østerby R, Parving H-H, Nyberg G, et al. A strong correlation between filtration rate and filtration surface in diabetic nephropathy. *Diabetologia.* 1988;31:265-270.
34. Harris RD, Steffes MW, Bilous RW, et al. Global glomerular sclerosis and glomerular arteriolar hyalinosis in insulin dependent diabetics. *Kidney Int.* 1991;40:107-144.
35. Mauer SM, Miller K, Goetz FC, et al. Immunopathology of renal extracellular membranes in kidneys transplanted into patients with diabetes mellitus. *Diabetes.* 1976;25:709-712.
36. Barbosa J, Johnson S. The Minnesota Diabetes Complications Clinical Trial: severe hypoglycemia during maximized insulin treatment of diabetes in a randomized clinical trial. *Diabetes Care.* 1983;6: 62-63.
37. Hung J, Menth L, Thompson MJ, et al. Minnesota Diabetes Complications Clinical Trial: cognitive functions under long-term maximized and standard metabolic controls. *Diabete Metab.* 1984;10: 48-55.
38. Chavers BM, Bilous RW, Ellis EW, et al. Glomerular lesions and urinary albumin excretion in type 1 diabetes without overt proteinuria. *N Engl J Med.* 1989;320:966-970.
39. Fioretto P, Steffes MW, Chavers BM, Mauer SM. Glomerular structure and function and blood pressure in normo- and micro albuminuric insulin dependent diabetic patients. *J Am Soc Nephrol.* 1992;3:332. Abstract.

Downloaded From: http://jama.jamanetwork.com/ by a Simon Fraser University User on 06/12/2015